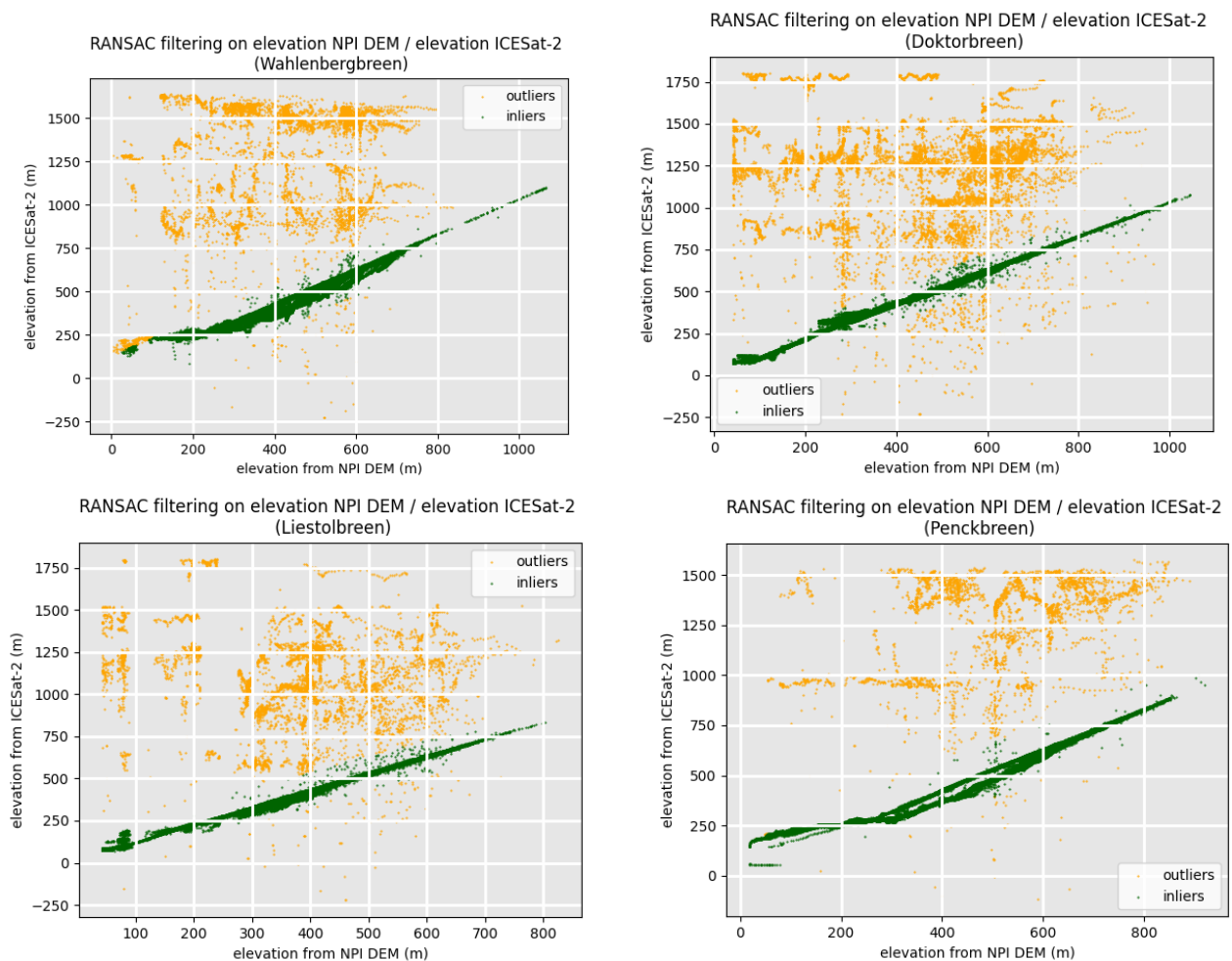
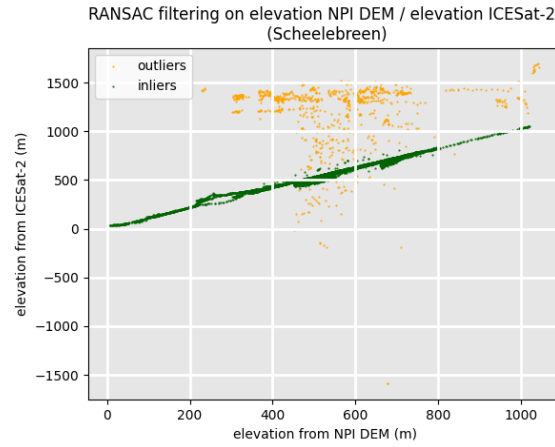


1. Filtering Using RANDOM SAmple Consensus (RANSAC) Approach.

The ICESat-2 elevation data included many outliers, mostly caused by cloud cover. However, elevation changes from glacier surges can also look like outliers. Because of this, a simple thresholding method like removing all points where the elevation difference (Δh) is above a certain value would likely remove both the unwanted cloud points and the critical surge signals. To help separate these different types of data, we compared ICESat-2 elevation with elevation from a digital elevation model (DEM). When plotted, this shows three groups: one that follows the glacier surface, one that is much higher (clouds), and one that is slightly offset (possible surge signals).

We needed a method that worked well across many different glaciers. For this, we used the RANSAC method to detect and remove outliers. RANSAC works by finding a line that fits the main glacier surface points, while ignoring those that don't fit (like clouds and surge points). Points within a threshold distance from this line were considered inliers. This allowed us to clean the data in a way that kept real glacier changes while removing most of the unwanted noise. The figures below show how RANSAC was used and how it improved the data quality. The method itself generally worked, however, it is important to note that some glaciers still contained a lot of noise, while on some very pronounced surges, some parts of the surge were removed (Wahlenbergreen). It was, however, the most effective method for the whole dataset.





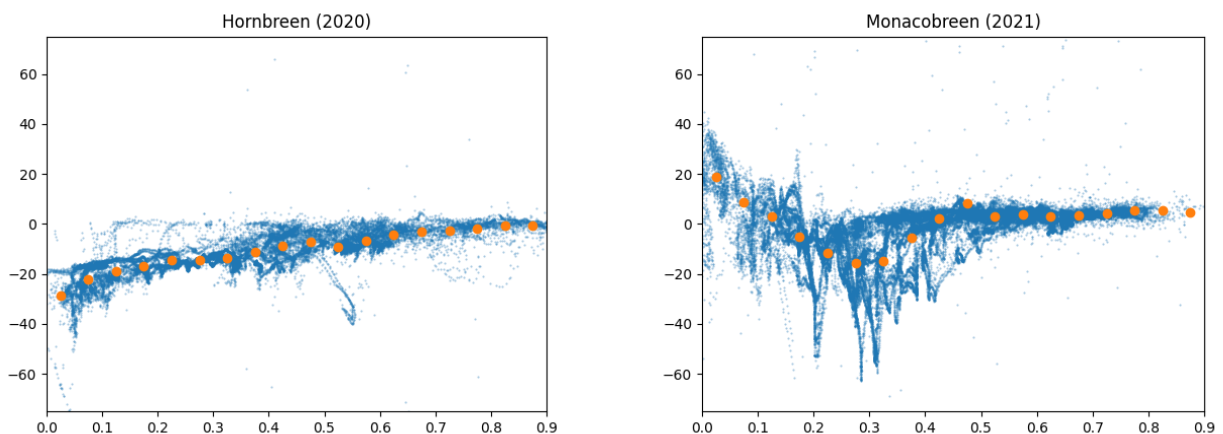
2. Feature Extraction

The features that were used for the Random Forest classification were briefly introduced in the main article. This supplementary section provides a more detailed explanation of all statistical features that were extracted, considered, and the motivation behind their use.

As established, a surge plot displays DEM-derived elevation on the x-axis and elevation change (dh) on the y-axis. In non-surging glaciers, these plots typically show consistent patterns: slight elevation gain in the accumulation zone and slight elevation loss in the ablation zone. Surging glaciers, on the other hand, exhibit chaotic behaviour – significant elevation gain in the lower parts and corresponding elevation loss in the upper parts, with no consistent magnitude.

The elevation data were normalized prior to analysis. Glaciers were divided into elevation bins, each representing 5% of the glacier's elevation range based on the DEM. This binning serves two purposes: to (1) ensure each part of the glacier contributes equally to the classification, avoiding dominance by high-elevation zones (which typically contain more data points) and (2) to allow for more consistent comparisons across different years. The lowest and highest elevation bins were excluded due to high error rates.

To identify surge-type glaciers using elevation change data we selected a set of statistical features designed to capture the distinct patterns associated with glacier surges. These features emphasize how both elevation and elevation change vary, with particular attention to the lower portion of the glacier, where surge-related surface changes are typically most pronounced. The following section details each feature, the rationale behind its inclusion, and how its values tend to differ between surging and non-surging glaciers.



95th percentile of elevation change in the lower part (dh_{max}). The 95th percentile highlights the extreme positive changes. Surging glaciers often show large increases in surface elevation due to ice pushing into the

lower glacier. The motivation for using this metric is that it should differentiate between surging and non-surging glaciers by a slightly negative value for non-surging glaciers, and a largely positive value for surging glaciers. The histogram shows this general trend. Nearly all non-surging glaciers stay near or below 0, while surging glaciers can have extreme positive values. There are some negative values for the surging glaciers as well. This might be caused between the difference between the acquisition time of the DEM and ICESat-2, which came well over 10 years later. It is possible that some glaciers lost a lot of mass after the DEM acquisition, then surged, gained mass, but still not enough to account for the mass loss in between the DEM acquisition and the surge.

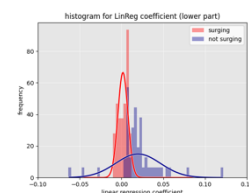
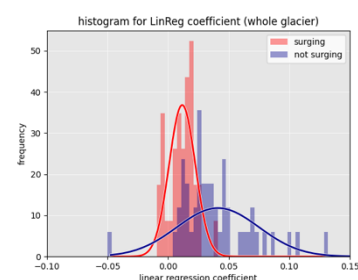
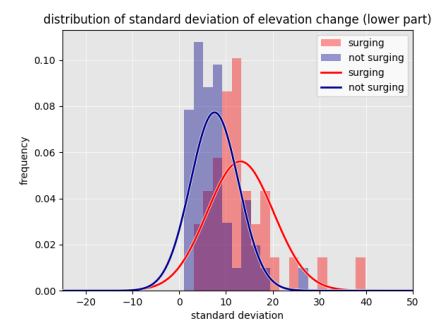
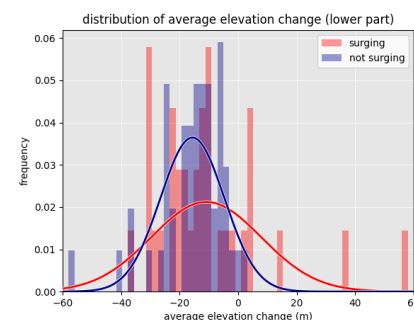
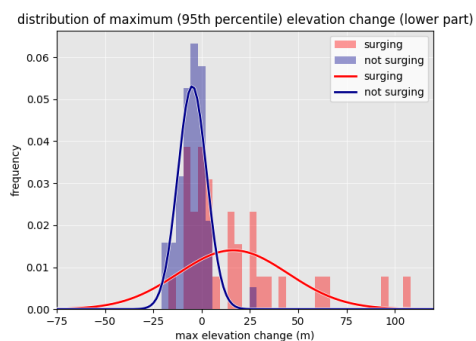
Mean elevation change in the lower part of the glacier (dh_mean_l). This one is similar to the 95th percentile, however, it is more focused on a “trend” than a single value. The idea is similar though – this metric should be negative for non-surging glaciers and positive for surging glaciers. This one is, however, more complicated and the values largely overlap. Even on a surging glacier, parts of the ablation zone may be experiencing elevation loss, pushing the average to negative values.

Standard deviation of elevation change in the lower part of the glacier (std_dev). This value might tell us about how much the elevation change is scattered, how large the difference is between the highest and lowest values. A stable glacier should have a low standard deviation, because the elevation change should be slightly negative throughout the ablation zone. Surging glaciers should have elevation changes in the ablation zone scattered all over the place, therefore showing a higher standard deviation. The distribution plot does not show dramatic differences between the categories.

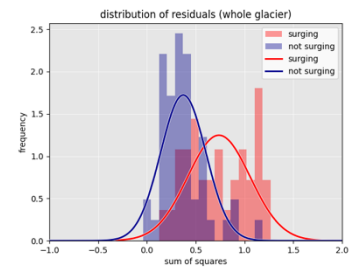
2D standard deviation of elevation and elevation change for the lower part of the glacier (hdh_stddev_l). The 2D standard deviation was used to compute standard deviation for x = elevation and y = elevation change. The motivation was to see if the values spread in one direction or are more scattered all over the place indicating a surge).

Linear regression coefficient on the binned data ($linreg_coef_binned$). The slope of the data was computed on the binned data using a linear regression. Typically, stable glaciers will have a positive slope (as explained before – accumulation in accumulation zone and ablation in ablation zone). Surging glaciers will have this completely the other way around as mass is transported from the upper parts towards the terminus. The slope was computed on the binned data to give the same weight to the narrow tongue (which contains less points) and the accumulation zone which is typically wider (contains more points).

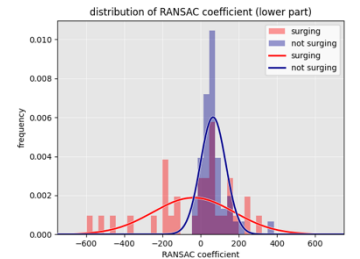
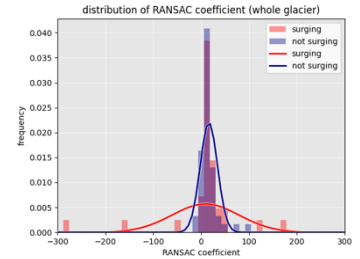
Linear regression coefficient on the binned data in the lower part of the glacier ($linreg_coef_l_binned$). The same was done for the lower part of the glacier, where surges are most pronounced.



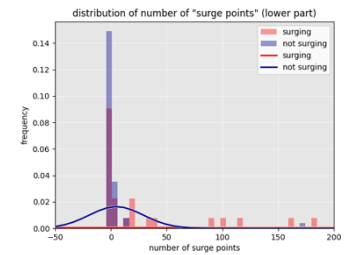
Residuals of linear regression (*residuals*). The residuals are computed as the sum of squares of the linear regression on the binned data. This metric is supposed to say how far the data is scattered from the line. Generally, non-surging glaciers should be close to the line, if the data does not contain noise. On surging glaciers, however, the points should be more scattered all over the place (it is likely that a linear regression is not the optimal regression to fit to the data), resulting in higher residuals.



RANSAC coefficient (*ransac*). The RANSAC regression coefficient was computed in order to calculate the trend of the data without potential noise. This was not computed on the binned data but on the actual ICESat-2 data points, to see the general trend of the elevation change per elevation. Typically, normal glaciers that are melting will have a slightly positive RANSAC coefficient when the glacier has lost mass in the ablation zone and gained mass in the accumulation zone. A very prominent surge will have the opposite signature (large accumulation in the lower part and mass loss in the upper part of the glacier), causing the RANSAC coefficient to be very negative. It can be, however, problematic when the surge is labeled as noise – then the value of a surging glacier is similar to a non-surging one.

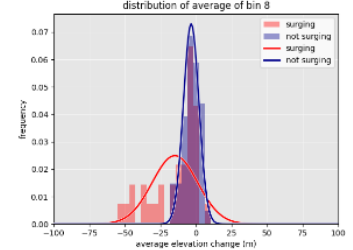
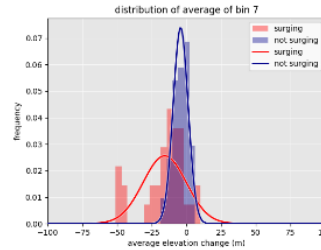
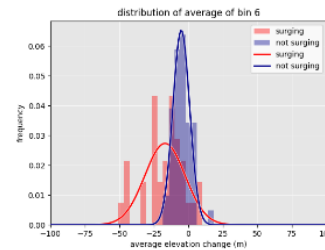
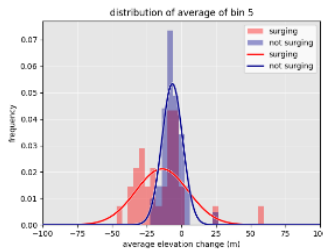
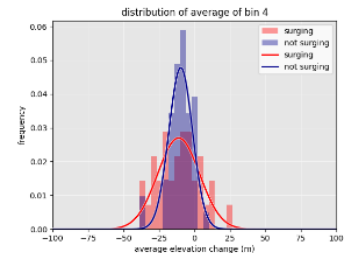
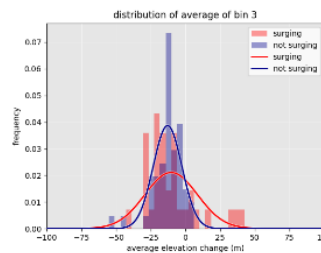
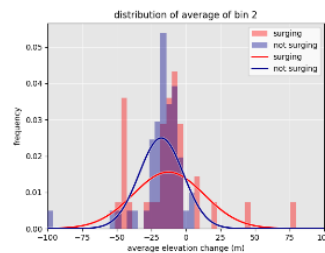
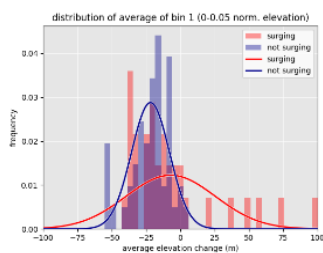


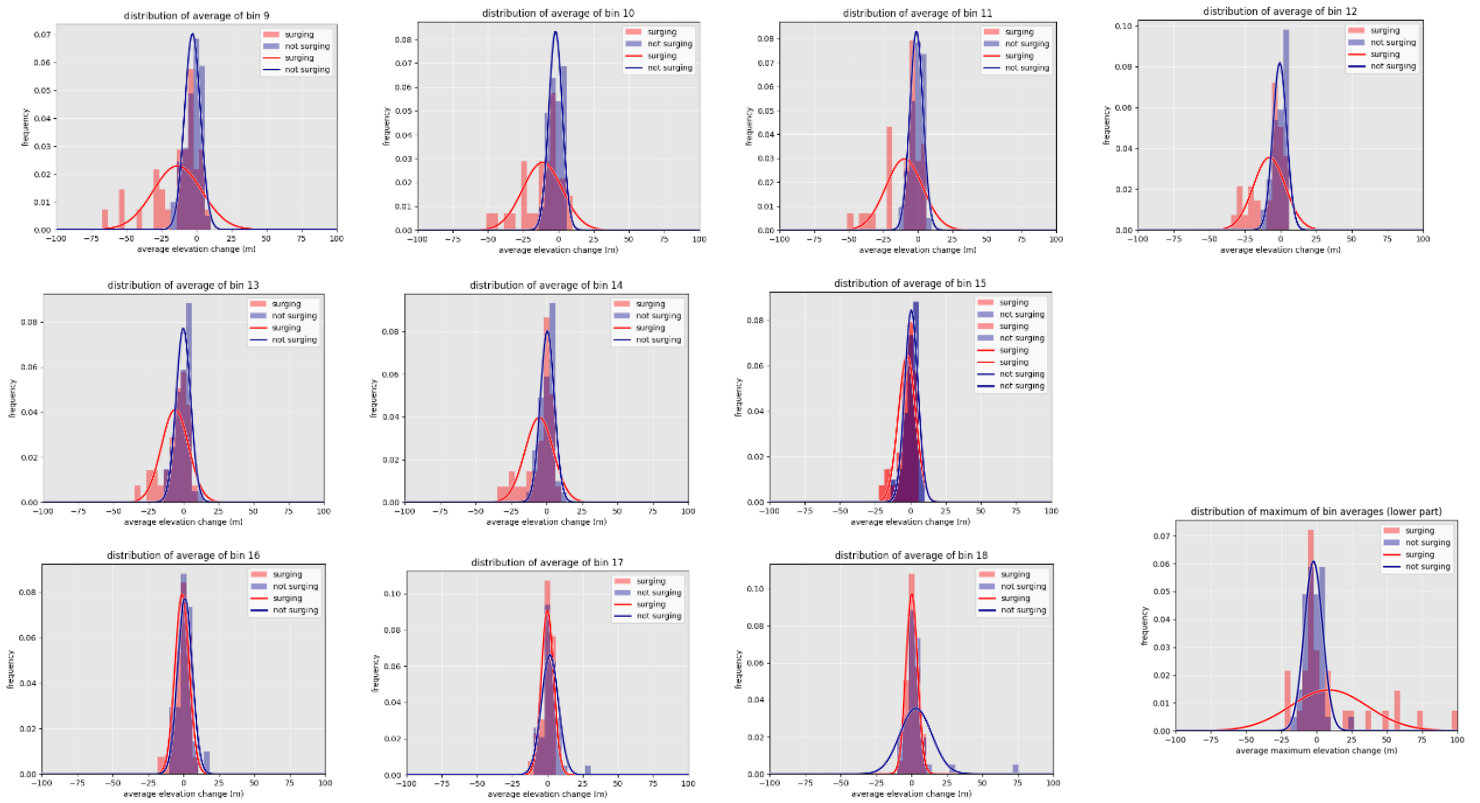
RANSAC coefficient for the lower part of the glacier (*ransac_l*).



Count of points above 15m elevation change in the lower part of the glacier (*over15*). These points should indicate a surge, however, could also be noise. The thought is that a surging glacier will have many more points above 15 m elevation change in the lower part, whereas if it is noise, it will just be isolated values. However, only the massive surges had many points over 15m, small surges did not have almost any.

Averages of the hypsometric bins (*bin1, ..., bin18*). The largest differences between surging and non-surging glaciers are visible in the lower bins (1 and 2) where surging glaciers often have larger average values and then in bins 5-14 where surging glaciers often have lower values than non-surging glaciers. In the upper-most bins there are no visible differences between the distributions.





Maximum elevation change value of the bins in the lower part of the glacier (*bin_max*). The maximum value from the bin averages in the lower part of the glacier is saved as a separate variable in order to enhance the influence of the bin where the surge is most pronounced.

3. Training Data

The training data for surging glaciers come from the inventory of Kääb et al. (2023) who identified surging glaciers between 2017 and 2023 based on SAR backscatter. Non-surging glaciers were manually selected.

training data for analysis for 5-year data			
	name	glacier_id	surging
0	Austfonna-Basin3	G024340E79634N	1
1	Arnesenbreen	G018098E77802N	1
2	Nathorstbreen	G016633E77290N	1
3	Sonklarbreen	G020098E78757N	1
4	Markhambreen	G017096E77164N	1
5	Recherchebreen	G015037E77377N	1
6	MorsjnevStrongbreen	G017697E77678N	1
7	Kvalbreen	G018031E77579N	1
8	Ganskijbreen	G020757E78746N	1
9	Negribreen	G018042E78675N	1
10	Tunabreen	G017497E78572N	1
11	Osbornebreen	G013139E78668N	1
12	Monacobreen	G012697E79319N	1
13	Stonebreen	G023608E77828N	1
14	Penckbreen	G015616E77394N	1
15	Scheelebreen	G016964E77694N	1
16	Vallakrabreen	G017158E77876N	1
17	Liestolbreen	G016915E77433N	1

18	Svalisbreen	G016777E76955N	1
19	Midtbreen	G016346E79525N	1
20	Bakaninbreen	G017525E7773N	0
21	Andrebreen	G011158E79339N	0
22	Mayerbreen	G012572E79241N	0
23	Kollerbreen	G012352E79323N	0
24	Elnabreen	G014132E79125N	0
25	Sjetdebreen	G011294E79478N	0
26	Karlsbreen	G013347E79296N	0
27	FjortendeJulibreen	G012363E79148N	0
28	Holmstrombreen	G013749E78857N	0
29	Lisbetbreen	G014909E79069N	0
30	VonPostbreen	G017824E78480N	0
31	VestreTorrellbreen	G015026E77342N	0
32	Steenstrupbreen	G016290E77607N	0
33	Inglefieldbreen	G017911E77804N	0
34	Andrinebreen	G018010E77975N	0
35	Jinnbreen	G017562E78141N	0
36	Hinlopenbreen	G018391E78924N	0
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38	Tryggvebreen	G016733E79123N	0
39	Marsjobreen	G022215E77928N	0
40	Kvitisen	G021454E78071N	0
41	Duckwitzbreen	G021120E78374N	0
42	Freemanbreen	G021590E78354N	0
43	VegafonnaMariebreen	G021009E79425N	0
44	Usherbreen	G018580E78256N	0
45	Nordmannsfonna	G018296E78282N	0
46	Hockstatterbreen	G019758E78902N	0
47	Tommelbreen	G018473E79506N	0
48	Svitdjodbreen	G011686E79700N	0
49	Kuhrbreen	G021849E77617N	0
50	Vitkovskibreen	G016292E76781N	0
51	Slakbreen	G016584E77967N	0
52	Dunerbreen	G016909E79728N	0
53	Morabreen	G014184E78936N	0
54	Forstebreen	G011244E79295N	0
55	Waggonwaybreen	G011437E79514N	0
56	Smerenburgbreen	G011600E79552N	0
57	Chauveaubreen	G011912E79647N	0
58	Raudfjordbreen	G012020E79575N	0

training data for analysis done by hydrological year				
	name	glacier_id	year	surging
0	Austfonna-Basin3	G024340E79634N	2019	1
1	Austfonna-Basin3	G024340E79634N	2020	1
2	Austfonna-Basin3	G024340E79634N	2021	1
3	Austfonna-Basin3	G024340E79634N	2022	1

4	Austfonna-Basin3	G024340E79634N	2023	1
5	Arnesenbreen	G018098E77802N	2019	1
6	Arnesenbreen	G018098E77802N	2020	1
7	Arnesenbreen	G018098E77802N	2023	1
8	Nathorstbreen	G016633E77290N	2019	1
9	Nathorstbreen	G016633E77290N	2020	1
10	Nathorstbreen	G016633E77290N	2021	1
11	Nathorstbreen	G016633E77290N	2022	1
12	Nathorstbreen	G016633E77290N	2023	1
13	Sonklarbreen	G020098E78757N	2020	1
14	Sonklarbreen	G020098E78757N	2021	1
15	Sonklarbreen	G020098E78757N	2022	1
16	Sonklarbreen	G020098E78757N	2023	1
17	Markhambreen	G017096E77164N	2019	1
18	Markhambreen	G017096E77164N	2020	1
19	Markhambreen	G017096E77164N	2021	1
20	Markhambreen	G017096E77164N	2023	1
21	Recherchebreen	G015037E77377N	2019	1
22	Recherchebreen	G015037E77377N	2020	1
23	Recherchebreen	G015037E77377N	2021	1
24	Recherchebreen	G015037E77377N	2022	1
25	Recherchebreen	G015037E77377N	2023	1
26	MorsjnevStrongbreen	G017697E77678N	2020	1
27	MorsjnevStrongbreen	G017697E77678N	2021	1
28	MorsjnevStrongbreen	G017697E77678N	2022	1
29	MorsjnevStrongbreen	G017697E77678N	2023	1
30	Kvalbreen	G018031E77579N	2020	1
31	Ganskijbreen	G020757E78746N	2019	1
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36	Negribreen	G018042E78675N	2022	1
37	Negribreen	G018042E78675N	2023	1
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41	Tunabreen	G017497E78572N	2022	1
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52	Stonebreen	G023608E77828N	2019	1
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62	Scheelebreen	G016964E77694N	2021	1
63	Vallakrabreen	G017158E77876N	2023	1
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65	Svalisbreen	G016777E76955N	2019	1
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69	Svalisbreen	G016777E76955N	2023	1
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95	Holmstrombreen	G013749E78857N	2019	0
96	Lisbetbreen	G014909E79069N	2019	0
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133	Nordmannsfonna	G018296E78282N	2019	0
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136	Tommelbreen	G018473E79506N	2021	0
137	Svitdjodbreen	G011686E79700N	2021	0
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145	Slakbreen	G016584E77967N	2021	0
146	Slakbreen	G016584E77967N	2023	0
147	Dunerbreen	G016909E79728N	2019	0

148	Dunerbreen	G016909E79728N	2020	0
149	Dunerbreen	G016909E79728N	2021	0
150	Dunerbreen	G016909E79728N	2022	0
151	Dunerbreen	G016909E79728N	2023	0
152	Morabreen	G014184E78936N	2021	0
153	Morabreen	G014184E78936N	2022	0
154	Morabreen	G014184E78936N	2023	0
155	Forstebreen	G011244E79295N	2019	0
156	Forstebreen	G011244E79295N	2021	0
157	Forstebreen	G011244E79295N	2022	0
158	Forstebreen	G011244E79295N	2023	0
159	Waggonwaybreen	G011437E79514N	2022	0
160	Waggonwaybreen	G011437E79514N	2023	0
161	Smerenburgbreen	G011600E79552N	2020	0
162	Smerenburgbreen	G011600E79552N	2021	0
163	Smerenburgbreen	G011600E79552N	2022	0
164	Smerenburgbreen	G011600E79552N	2023	0
165	Chauveaubreen	G011912E79647N	2020	0
166	Raudfjordbreen	G012020E79575N	2020	0
167	Raudfjordbreen	G012020E79575N	2021	0

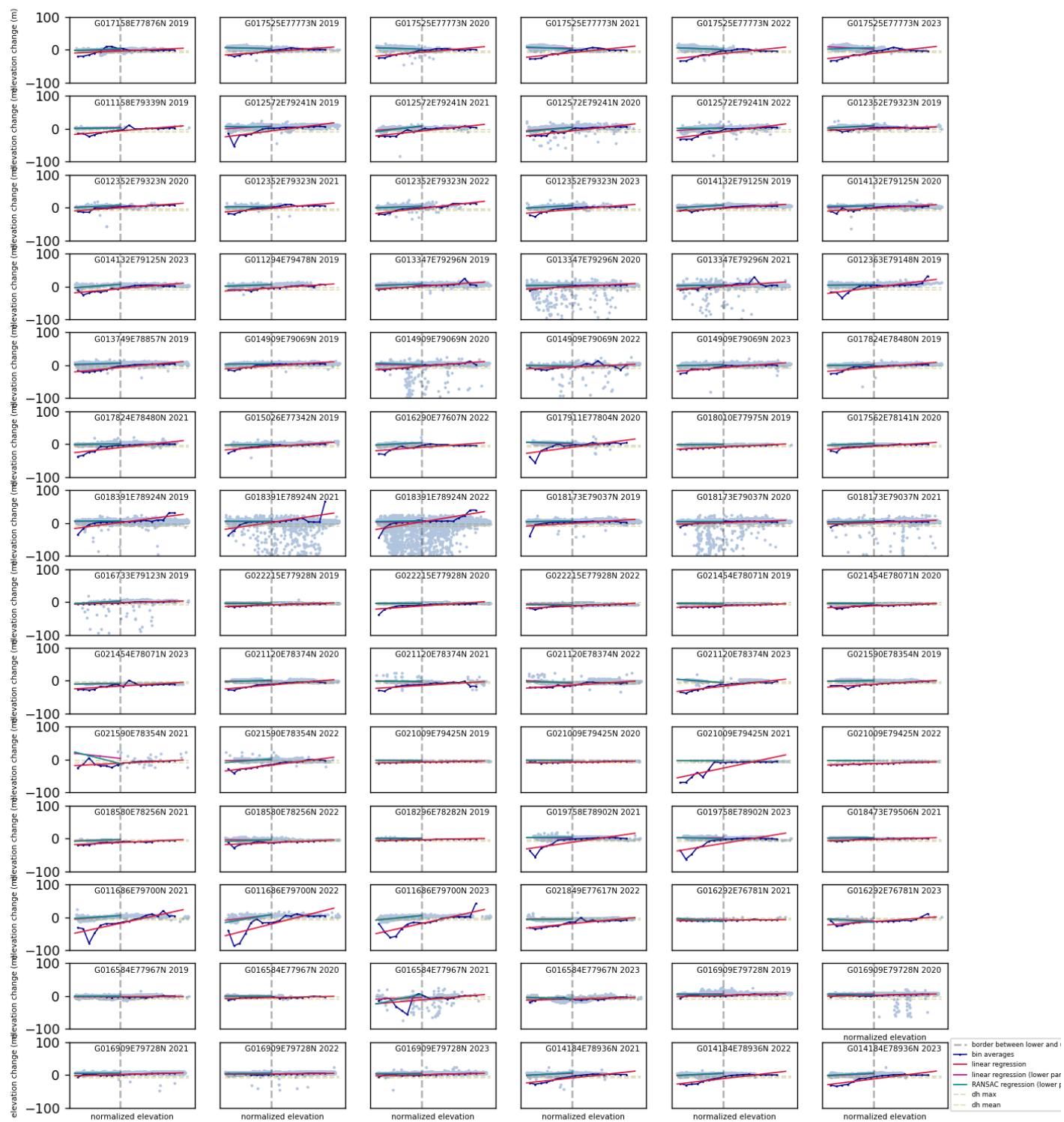


Figure 1 - Features of non-surging glaciers.

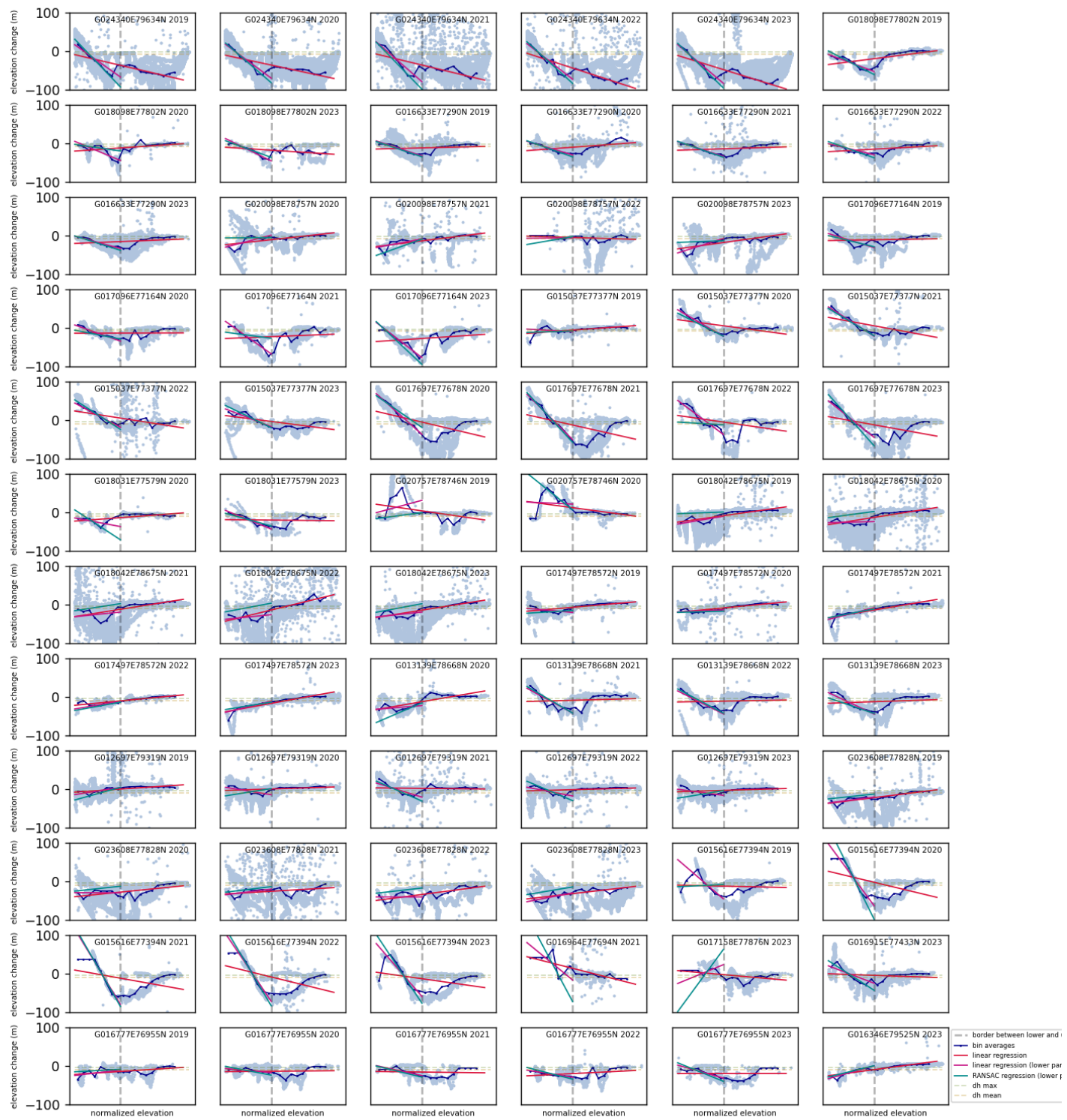


Figure 2 - Features of surging glaciers.

4. Results (and comparison with other studies)

Here I present the full list of glaciers classified as surging in a comprehensive table including glacier name, probability with which it was classified as surging, whether or not the glacier was included in the training data column “training data”), results of manual validation (column “validation”). Manual validation included several steps: (1) cross-checking literature and searching for mentions of a surge in existing studies, (2) visual evaluation of sudden terminus advance, and/or sudden crevassing in optical satellite imagery (Sentinel-2), (3) visual evaluation of surface velocity from the ITS_LIVE dataset.

There are two tables of results: one set of results where the analysis was conducted on the whole ICESat-2 dataset (2018-2024), and one for the analysis where the data were grouped by hydrological year. Only glaciers classified as surging are presented in this supplementary file. The full results of the classification are available for download on GitHub on the link <https://github.com/eliskasieglova/SvalbardSurges/tree/main>. False negatives were manually added to this list (glaciers that we know were surging during the study period, but were not classified as surging).

Glaciers with a good quality flag classified as surging between DEM acquisition dates (2008-2012) and ICESat-2 data acquisition dates (2018-2024).						
glacier_id	glacier_name	surging	probability	training	validation	source
G017096E77164N	Markhambreen	1	0,959193788	1	1	Szafraniec (2020)
G013139E78668N	Osbornebreen	1	0,958527121	1	1	Guillet et al. (2025)
G015502E77425N	Sveitsarfonna	1	0,958127121		0	
G017697E77678N	Morsjnevbreen / Strongbreen	1	0,957038377	1	1	Guillet et al. (2025)
G016187E77275N	Polakkbreen	1	0,952232888		1	Guillet et al. (2025)
G016633E77290N	Nathorstbreen	1	0,952017099	1	1	Guillet et al. (2025)
G015616E77394N	Penckbreen	1	0,947590536	1	1	Guillet et al. (2025)
G018098E77802N	Arnesenbreen	1	0,94008487	1	1	Guillet et al. (2025)
G026458E79881N	Austfonna, -	1	0,928236329		1	Zheng (2022), Guillet et al. (2025)
G016777E76955N	Svalisbreen	1	0,920386432	1	1	
G016915E77433N	Liestølbreen	1	0,919591192	1	1	Guillet et al. (2025)
G024340E79634N	Austfonna, -	1	0,918892319	1	1	Zheng (2022), Guillet et al. (20245)
G024396E79406N	Austfonna, -	1	0,917392319		1	Zheng (2022)
G015037E77377N	Recherchebreen	1	0,901219284	1	1	Guillet et al. (2025)
G012186E79547N	Emmabreen	1	0,894625185		1	Alean and Hambrey (2017)
G012697E79319N	Monacobreen	1	0,894361225	1	1	Guillet et al. (2025)
G012307E78697N	Aavatsmarkbreen	1	0,892447956		1	Sevestre, (2018), Guillet et al. (2025)
G013747E78362N	Esmarkbreen	1	0,889021988		1	Guillet et al. (2025)
G018031E77579N	Kvalbreen	1	0,886555098	1	1	Guillet et al. (2025)
G013901E78579N	Wahlenbergbreen	1	0,885632152		1	Sevestre (2018), Guillet et al. (2025)
G016561E77390N		1	0,879756723		0	
G020757E78746N	Ganskijbreen	1	0,879673226	1	1	Guillet et al. (2025)
G016964E77694N	Scheelebreen	1	0,866093245	1	1	Guillet et al. (2025)

G012520E78787N	Uversbreen	1	0,856310751		1	Sysselmannen (2014))
G012468E79076N	Blomstrandbreen	1	0,839172303		1	Sund and Eiken (2017)
G016364E79743N	Åsgardfonna NW	1	0,831206277		2	
G017158E77876N	Vallåkrabreen	1	0,829818215	1	1	Guillet et al. (2025)
G016885E77574N	Doktorbreen	1	0,826908218		1	Guillet et al. (2025)
G016598E79263N	Cookbreen	1	0,819642423		2	
G023229E79872N	Austfonna, -	1	0,802853747		2	
G023016E77464N		1	0,786606514		0	
G023619E79932N	Austfonna, -	1	0,773007059		2	
G022184E77709N		1	0,752499629		0	
G022867E77386N		1	0,749538305		2	Sund and others (2014)
G017687E78107N	Sveigbreen	1	0,743522669		0	
G017731E78159N	Skruisbreen	1	0,723813945		2	
G014342E77910N		1	0,722904825		2	
G013819E78747N	Sefströmbreen	1	0,716166626		1	Guillet et al. (2025)
G017583E78369N	Fimbulisen W	1	0,713358614		0	
G022129E77665N	Veidebreen	1	0,712205038		0	
G015857E77338N	Zawadzki breen	1	0,710564221		1	Guillet et al. (2025)
G020164E80193N		1	0,709752189		2	
G017399E77693N	Paulabreen	1	0,705825134		1	Guillet et al. (2025)
G014869E79036N		1	0,704128467		0	
G016898E76901N	Vasilievbreen	1	0,701839463		2	Guillet et al. (2025)
G023608E77828N	Stonebreen	1	0,700821977	1	1	Koch et al. (2023), Guillet et al. (2025)
G021526E77589N		1	0,697690324		0	
G025164E79993N	Austfonna, Nilsenbreen	1	0,696125857		2	
G018199E78191N	Elfenbeinbreen	1	0,694720415		2	
G022338E77853N		1	0,684321502		0	
G022417E77913N		1	0,659397301		0	
G015900E77167N	Mülbacherbreen	1	0,657142543		0	
G014532E79489N		1	0,656048112		0	
G024143E79973N	Austfonna, Duvebreen	1	0,655638869		2	
G016482E76791N	Olsokbreen	1	0,644728393		2	
G022918E77824N	Seidbreen	1	0,643655466		0	
G016945E78009N		1	0,639420226		2	
G014941E78875N		1	0,635935427		0	
G017693E79137N	Chydeniusbreen	1	0,627850541		2	
G016346E79525N	Midtbreen	1	0,62365707	1	1	Guillet et al. (2025)
G016634E77343N	Dobrowolski breen	1	0,613907333		1	Guillet et al. (2025)
G014063E78811N		1	0,603971818		0	
G014813E79143N		1	0,598831198		0	
G025456E80055N	Austfonna, Sexebreen	1	0,597563706		0	
G020098E78757N	Sonklarbreen	1	0,595226735	1	1	Guillet et al. (2025)

G018042E78675N	Negribreen	1	0,583897607	1	1	Guillet et al. (2025)
G017333E77537N	Nuddbreen / Strongbreen	1	0,580764461		1	Guillet et al. (2025)
G023592E77662N	Kong Johans Bre	1	0,571875658		0	
G023101E77888N	Albrechtbreen	1	0,564233239		0	
G021502E79897N	Bodleybreen	1	0,560065546		2	
G012787E79535N		1	0,559711433		0	
G013271E78658N	Osbornebreen	1	0,55779026		1	Guillet et al. (2025)
G011663E79770N		1	0,552435452		0	
G021338E77589N		1	0,552182346		0	
G019797E80009N	Søre Franklinbreen	1	0,547984401		1	Guillet et al. (2025)
G016557E76915N	Mendelejevbre	1	0,545485087		0	
G018751E79906N		1	0,545385129		0	
G017180E77206N	Crollbreen	1	0,545000022		1	Sentine-2 imagery
G022015E77578N		1	0,543219386		0	
G023386E77490N	Pettersenbreen	1	0,538613642		2	
G023062E77750N	Gandbreen N	1	0,535392038		0	
G013739E79134N	Vonbreen	1	0,526732436		2	

Glaciers classified as surging between DEM acquisition dates (2008-2012) and corresponding hydrological year in which ICESat-2 data was collected.						
glacier_id	glacier_name	year	surging	probability	training	validation
G012697E79319N	Monacobreen	2019	1	0,549511269	1	1
G012697E79319N	Monacobreen	2020	1	0,898073318	1	1
G012697E79319N	Monacobreen	2021	1	0,92295305	1	1
G012697E79319N	Monacobreen	2022	1	0,898226396	1	1
G012697E79319N	Monacobreen	2023	1	0,876780997	1	1
G013470E79201N	Vonbreen	2021	1	0,662668581		1
G012468E79076N	Blomstrandbreen	2019	1	0,817812475		1
G012468E79076N	Blomstrandbreen	2020	1	0,869637576		1
G012468E79076N	Blomstrandbreen	2021	1	0,8336825		1
G012468E79076N	Blomstrandbreen	2022	1	0,741201896		1
G012468E79076N	Blomstrandbreen	2023	1	0,853805232		1
G013248E78855N	Infantfonna	2023	1	0,65339193		1
G012520E78787N	Uversbreen	2019	1	0,824488079		1
G012520E78787N	Uversbreen	2020	1	0,814089904		1
G012520E78787N	Uversbreen	2021	1	0,817340274		1
G012520E78787N	Uversbreen	2022	1	0,795172155		1
G012520E78787N	Uversbreen	2023	1	0,704491137		1
G012307E78697N	Aavatsmarkbreen	2019	1	0,933559434		1
G012307E78697N	Aavatsmarkbreen	2020	1	0,95420834		1
G012307E78697N	Aavatsmarkbreen	2021	1	0,849083791		1
G012307E78697N	Aavatsmarkbreen	2022	1	0,913079623		1
G012307E78697N	Aavatsmarkbreen	2023	1	0,879526618		1
G012918E78620N	Konowbreen	2022	1	0,544652462		1

G012918E78620N	Konowbreen	2023	1	0,687266402		1
G012670E78658N	Dahlbreen	2021	1	0,527928209		1
G012670E78658N	Dahlbreen	2022	1	0,678066894		1
G012670E78658N	Dahlbreen	2023	1	0,677321719		1
G013139E78668N	Osbornebreen	2020	1	0,978970141	1	1
G013139E78668N	Osbornebreen	2021	1	0,992434028	1	1
G013139E78668N	Osbornebreen	2022	1	0,999777778	1	1
G013139E78668N	Osbornebreen	2023	1	0,999492063	1	1
G013271E78658N	Osbornebreen	2021	1	0,532622317		1
G013271E78658N	Osbornebreen	2022	1	0,619042745		1
G013747E78362N	Esmarkbreen	2020	1	0,74113475		1
G013747E78362N	Esmarkbreen	2021	1	0,829540886		1
G013747E78362N	Esmarkbreen	2022	1	0,799668678		1
G013747E78362N	Esmarkbreen	2023	1	0,808670851		1
G013901E78579N	Wahlenbergbreen	2019	1	0,974882679		1
G013901E78579N	Wahlenbergbreen	2020	1	0,950430229		1
G013901E78579N	Wahlenbergbreen	2021	1	0,959845499		1
G013901E78579N	Wahlenbergbreen	2022	1	0,989979874		1
G013901E78579N	Wahlenbergbreen	2023	1	0,994469571		1
G014909E79069N	Lisbetbreen	2021	1	0,643432897		1
G017497E78572N	Tunabreen	2019	1	0,633540774	1	1
G017497E78572N	Tunabreen	2020	1	0,704142469	1	1
G017497E78572N	Tunabreen	2022	1	0,610733821	1	1
G014257E77913N	Vestre Gronfjordbreen	2023	1	0,646104525		1
G014123E77878N		2023	1	0,560079861		1
G017399E77693N	Paulabreen	2022	1	0,529661311		1
G017399E77693N	Paulabreen	2023	1	0,524265833		1
G017697E77678N	Morsjnevbreenn / Strongbreen	2019	1	0,957461871		1
G017697E77678N	Morsjnevbreenn / Strongbreen	2020	1	0,975111765	1	1
G017697E77678N	Morsjnevbreenn / Strongbreen	2021	1	0,986061637	1	1
G017697E77678N	Morsjnevbreenn / Strongbreen	2022	1	0,978934327	1	1
G017697E77678N	Morsjnevbreenn / Strongbreen	2023	1	0,965485418	1	1
G018031E77579N	Kvalbreen	2019	1	0,595867175		1
G018031E77579N	Kvalbreen	2020	1	0,70628865	1	1
G018031E77579N	Kvalbreen	2021	1	0,997574224		1
G017096E77164N	Markhambreen	2019	1	1	1	1
G017096E77164N	Markhambreen	2020	1	0,997681852	1	1
G017096E77164N	Markhambreen	2021	1	0,943962078	1	1
G017096E77164N	Markhambreen	2023	1	0,986349838	1	1
G016898E76901N	Vasilievbreen	2019	1	0,91295678		1
G016898E76901N	Vasilievbreen	2020	1	0,84495343		1
G016898E76901N	Vasilievbreen	2022	1	0,625647845		1

G016898E76901N	Vasilievbreen	2023	1	0,893926498		1
G016777E76955N	Svalisbreen	2019	1	0,573483778	1	1
G016777E76955N	Svalisbreen	2020	1	0,878722998	1	1
G016777E76955N	Svalisbreen	2021	1	0,914737421	1	1
G016777E76955N	Svalisbreen	2022	1	0,971424038	1	1
G016777E76955N	Svalisbreen	2023	1	0,987689196	1	1
G016557E76915N	Mendelevbreen	2020	1	0,67440432		1
G016557E76915N	Mendelevbreen	2021	1	0,639107142		1
G016557E76915N	Mendelevbreen	2022	1	0,640141382		1
G016557E76915N	Mendelevbreen	2023	1	0,676183824		1
G016482E76791N	Olsokbreen	2019	1	0,59365888	0	1
G016482E76791N	Olsokbreen	2020	1	0,732813907		1
G016482E76791N	Olsokbreen	2021	1	0,689723314		1
G016482E76791N	Olsokbreen	2022	1	0,839659723		1
G016482E76791N	Olsokbreen	2023	1	0,855238271		1
G016757E76881N	Vasilievbreen	2021	1	0,79275196		1
G016757E76881N	Vasilievbreen	2022	1	0,79487731		1
G016757E76881N	Vasilievbreen	2023	1	0,54664602		1
G015037E77377N	Recherchebreen	2019	1	0,623019315	1	1
G015037E77377N	Recherchebreen	2020	1	0,923352171	1	1
G015037E77377N	Recherchebreen	2021	1	0,856886873	1	1
G015037E77377N	Recherchebreen	2022	1	0,899383625	1	1
G015037E77377N	Recherchebreen	2023	1	0,929645197	1	1
G015502E77425N	Sveitsarfonna	2019	1	0,975029179		1
G015502E77425N	Sveitsarfonna	2020	1	0,663268884		1
G015616E77394N	Penckbreen	2019	1	0,992874074	1	1
G015616E77394N	Penckbreen	2020	1	0,990780537	1	1
G015616E77394N	Penckbreen	2021	1	0,992085299	1	1
G015616E77394N	Penckbreen	2022	1	0,989474982	1	1
G015616E77394N	Penckbreen	2023	1	0,992344372	1	1
G015857E77338N	Zawadzkiibreen	2019	1	0,84865367		1
G015857E77338N	Zawadzkiibreen	2020	1	0,858578457		1
G015857E77338N	Zawadzkiibreen	2021	1	0,86551023		1
G015857E77338N	Zawadzkiibreen	2022	1	0,822296004		1
G015857E77338N	Zawadzkiibreen	2023	1	0,823743675		1
G016187E77275N	Polakkbreen	2019	1	0,990243515		1
G016187E77275N	Polakkbreen	2020	1	0,928044286		1
G016187E77275N	Polakkbreen	2021	1	0,988655091		1
G016187E77275N	Polakkbreen	2022	1	0,988655091		1
G016187E77275N	Polakkbreen	2023	1	0,982777948		1
G016915E77433N	Liestølbreen	2022	1	0,925136447		1
G016915E77433N	Liestølbreen	2023	1	0,96892459	1	1
G016634E77343N	Dobrowolskiibreen	2019	1	0,721811174		1
G016634E77343N	Dobrowolskiibreen	2020	1	0,719799447		1
G016634E77343N	Dobrowolskiibreen	2021	1	0,715272214		1
G016634E77343N	Dobrowolskiibreen	2022	1	0,730610889		1

G016634E77343N	Dobrowolskibreen	2023	1	0,68343662		1
G016633E77290N	Nathorstbreen	2019	1	0,99776473	1	1
G016633E77290N	Nathorstbreen	2020	1	0,992999192	1	1
G016633E77290N	Nathorstbreen	2021	1	0,969880338	1	1
G016633E77290N	Nathorstbreen	2022	1	0,98665021	1	1
G016633E77290N	Nathorstbreen	2023	1	0,973880621	1	1
G017333E77537N	Nuddbreen / Strongbreen	2023	1	0,53180475		1
G016885E77574N	Doktorbreen	2021	1	0,542983261		1
G016885E77574N	Doktorbreen	2022	1	0,674806345		1
G016885E77574N	Doktorbreen	2023	1	0,656294153		1
G018098E77802N	Arnesenbreen	2019	1	0,968271232	1	1
G018231E77758N	Ingerbreen	2021	1	0,615295895		1
G018079E77679N	Innifonna	2020	1	0,527827208		1
G017893E77712N	Richardsbreen	2019	1	0,529352183		1
G017893E77712N	Richardsbreen	2020	1	0,539271303		1
G017025E78797N	Mittag-Lefflerbreen	2022	1	0,645896499		1
G017693E79137N	Chydeniusbreen	2022	1	0,529887339		1
G017730E79297N	Kantbreen	2020	1	0,536431897		1
G020098E78757N	Sonklarbreen	2021	1	0,619226234	1	1
G020098E78757N	Sonklarbreen	2022	1	0,610317523	1	1
G020098E78757N	Sonklarbreen	2023	1	0,625790394	1	1
G020757E78746N	Ganskijbreen	2019	1	0,715762863	1	1
G020757E78746N	Ganskijbreen	2020	1	0,684403037	1	1
G020757E78746N	Ganskijbreen	2021	1	0,76856908		1
G018419E78498N	Petermannbreen	2020	1	0,574764724		1
G018419E78498N	Petermannbreen	2022	1	0,711132843		1
G018419E78498N	Petermannbreen	2023	1	0,762087073		1
G018042E78675N	Negribreen	2019	1	0,640068334	1	1
G018042E78675N	Negribreen	2020	1	0,84437305	1	1
G018042E78675N	Negribreen	2021	1	0,863109809	1	1
G018042E78675N	Negribreen	2022	1	0,856662954	1	1
G018042E78675N	Negribreen	2023	1	0,788757609	1	1
G013819E78747N	Sefströmbreen	2021	1	0,557268094		1
G013819E78747N	Sefströmbreen	2022	1	0,584208016		1
G013819E78747N	Sefströmbreen	2023	1	0,589375171		1
G012186E79547N	Emmabreen	2019	1	0,774145657		1
G012186E79547N	Emmabreen	2023	1	0,870649428		1
G011600E79552N	Smeerenburgbreen	2023	1	0,570335244	0	1
G011686E79700N	Svitjodbreen	2019	1	0,545708948		1
G011686E79700N	Svitjodbreen	2021	1	0,565158742	0	1
G011686E79700N	Svitjodbreen	2023	1	0,585306083	0	1
G011663E79770N		2019	1	0,54832612		1
G011860E79450N	Lilliehøøkbreen	2023	1	0,808030027		1
G019797E80009N	Søre Franklinbreen	2020	1	0,542090228		1
G019797E80009N	Søre Franklinbreen	2022	1	0,589568223		1
G020164E80193N		2020	1	0,551566502		1

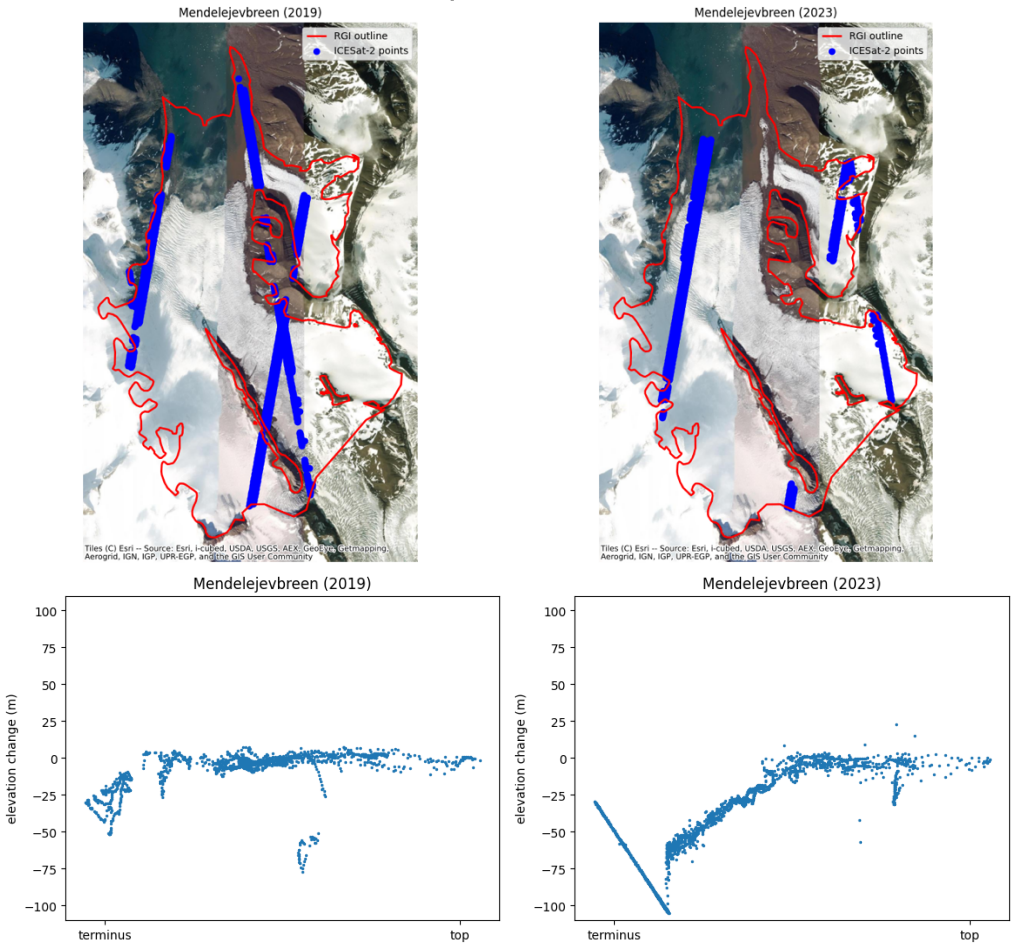
G021502E79897N	Bodleybreen	2021	1	0,614050571		1
G021502E79897N	Bodleybreen	2023	1	0,528525813		1
G026669E79935N	Austfonna, -	2019	1	0,604836919		1
G026669E79935N	Austfonna, -	2022	1	0,528096493		1
G026458E79881N	Austfonna, -	2019	1	0,575709862		1
G026458E79881N	Austfonna, -	2020	1	0,940425664		1
G026458E79881N	Austfonna, -	2021	1	0,954369616		1
G026458E79881N	Austfonna, -	2022	1	0,894721814		1
G026458E79881N	Austfonna, -	2023	1	0,953311123		1
G024340E79634N	Austfonna, -	2019	1	0,974999256	1	1
G024340E79634N	Austfonna, -	2020	1	0,973392527	1	1
G024340E79634N	Austfonna, -	2021	1	0,97587914	1	1
G024340E79634N	Austfonna, -	2022	1	0,965288663	1	1
G024340E79634N	Austfonna, -	2023	1	0,952000067	1	1
G024396E79406N	Austfonna, -	2019	1	0,943843687		1
G024396E79406N	Austfonna, -	2020	1	0,937586066		1
G024396E79406N	Austfonna, -	2021	1	0,950273979		1
G024396E79406N	Austfonna, -	2022	1	0,952441337		1
G024396E79406N	Austfonna, -	2023	1	0,952978133		1
G021590E78354N	Freemanbreen	2023	1	0,538275568		1
G021460E78472N	Besselsbreen	2021	1	0,842936661		1
G021460E78472N	Besselsbreen	2022	1	0,63832209		1
G021460E78472N	Besselsbreen	2023	1	0,623598264		1
G023101E77888N	Albrechtbreen	2021	1	0,542404989		1
G023101E77888N	Albrechtbreen	2022	1	0,689273606		1
G023101E77888N	Albrechtbreen	2023	1	0,652428994		1
G022918E77824N	Seidbreen	2023	1	0,755905022		1
G023062E77750N	Gandbreen N	2020	1	0,575325321		1
G023062E77750N	Gandbreen N	2023	1	0,537317734		1
G023134E77724N	Gandbreen S	2019	1	0,56228827		1
G023294E77648N	Deltabreen	2020	1	0,59105002		1
G023294E77648N	Deltabreen	2021	1	0,565777377		1
G023294E77648N	Deltabreen	2022	1	0,528499316		1
G023294E77648N	Deltabreen	2023	1	0,546567837		1
G023592E77662N	Kong Johans Bre	2019	1	0,671458515		1
G023592E77662N	Kong Johans Bre	2020	1	0,702820278		1
G023592E77662N	Kong Johans Bre	2021	1	0,805007698		1
G023016E77464N		2019	1	0,594961718		1
G022867E77386N		2022	1	0,686201918		1
G022129E77665N	Veidebreen	2022	1	0,863098323		1
G022129E77665N	Veidebreen	2023	1	0,765465642		1
G021849E77617N	Kuhrbreen	2019	1	0,579355179		1
G021849E77617N	Kuhrbreen	2022	1	0,580602611	0	1
G021849E77617N	Kuhrbreen	2023	1	0,679436602		1
G023608E77828N	Stonebreen	2019	1	0,819960182	1	1
G023608E77828N	Stonebreen	2020	1	0,830624831	1	1

G023608E77828N	Stonebreen	2021	1	0,842079641	1	1
G023608E77828N	Stonebreen	2022	1	0,90862949	1	1
G023608E77828N	Stonebreen	2023	1	0,834336281	1	1
G015444E77229N	Austre Torellbreen	2022	1	0,703124235		1
G015444E77229N	Austre Torellbreen	2023	1	0,579584995		1
G015696E77195N	Paierlbreen	2022	1	0,547520591		1
G015696E77195N	Paierlbreen	2023	1	0,691499676		1
G015900E77167N	Mülbacherbreen	2020	1	0,649512664		1
G015900E77167N	Mülbacherbreen	2021	1	0,705648297		1
G015900E77167N	Mülbacherbreen	2022	1	0,660616971		1
G015900E77167N	Mülbacherbreen	2023	1	0,575254311		1
G016807E77171N	Hornbreen	2022	1	0,530174394		1
G013724E78479N	Borebreen	2023	1	0,725300448		1

5. Outdated RGI outlines causing false positives.

The RGI 7.0 glacier outlines were published in 2023 and are based on Landsat 7 ETM satellite imagery from the year 2002. Since then, many glaciers have retreated and the outlines are now outdated. This can cause multiple problems also in misclassification.

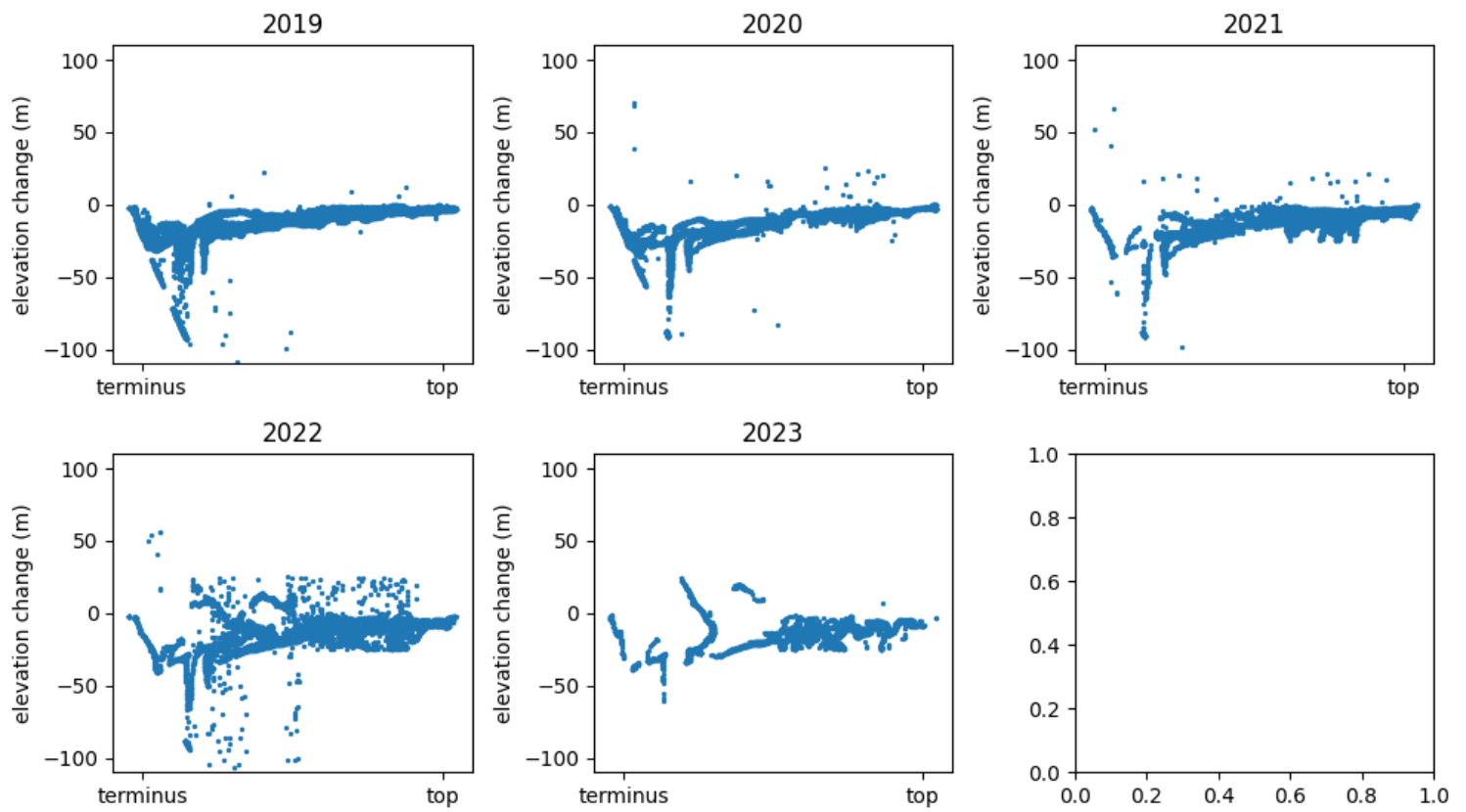
Mendelejevreen was classified as surging all years between except for 2019. In 2019 the tracks did not include data from the (now) fjord. This is what the plots for the other years (example on 2023) looked like:



6. Elevation change plots for a couple discussed glaciers

There is a visible surge bulge on Deltabreen in 2023 before it started surging in 2024.

Deltabreen



iefstrombreen also has a surge bulge visible and was classified as surging since 2021.

Sefströmbreen

